Granular Instrumental Variables

Estimating Aggregate Effects of Firm Shocks on Wages

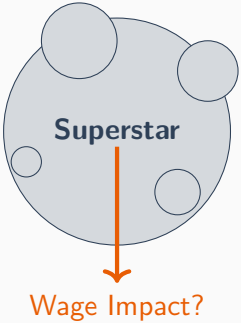
Macro Lab Group Presentation

Do shocks to superstar firms drive aggregate wage dynamics?

When a dominant employer experiences

an unexpected surge in productivity, local and aggregate wages often respond.

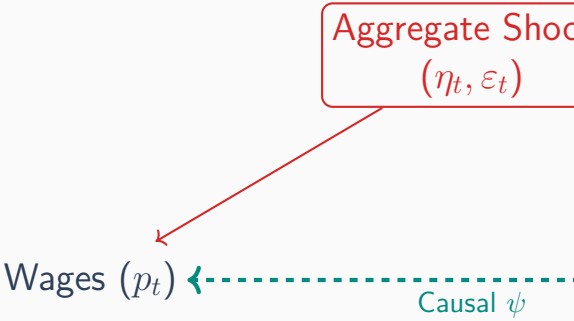
But how much of that response is causal, and how much is driven by common macroeconomic trends?



Endogenous aggregate shocks confound the effect of large firms

We want to estimate the elasticity of wages to labor demand.

The fundamental challenge:



A Simple Model

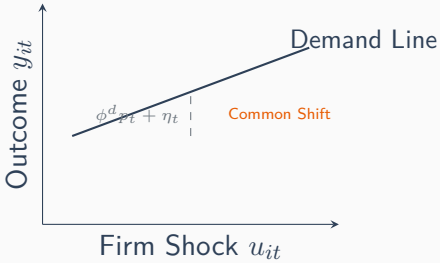
Section III.A of Gabaix and Koijen (2022)

A simple model of aggregate and idiosyncratic shocks

The system is given by:

$$p_t = \psi y_{St} + \varepsilon_t \tag{38}$$

$$y_{it} = \phi^d p_t + \eta_t + u_{it} \tag{39}$$



Where $y_{St} = \sum_i S_i y_{it}$ is the size-weighted aggregate outcome.

Constructing the Granular Instrumental Variable (z_t)

The GIV exploits the difference between size-weighted and equal-weighted averages.

$$\underbrace{y_{St}}_{\text{Size-Weighted}} - \underbrace{y_{Et}}_{\text{Equal-Weighted}} = \underbrace{z_t}_{\text{The GIV}} \qquad z_t = y_{St} - y_{Et}$$

The GIV isolates purely idiosyncratic firm shocks

$$y_{St} = \underbrace{\phi^d p_t + \eta_t}_{\text{Common}} + u_{St}$$

$$y_{Et} = \underbrace{\phi^d p_t + \eta_t}_{\text{Common}} + u_{Et}$$

Subtraction

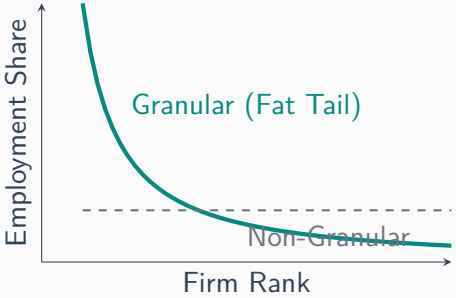
$$\tilde{y}_t = u_{St} - u_{Et}$$

Common factors vanish, leaving only idiosyncratic "grain"

GIV precision relies on concentrated labor markets

For the instrument to be strong, we need:

1. **A Large Excess Herfindahl (h):** Concentrated markets.
2. **Volatile Micro-Shocks (σ_u)**

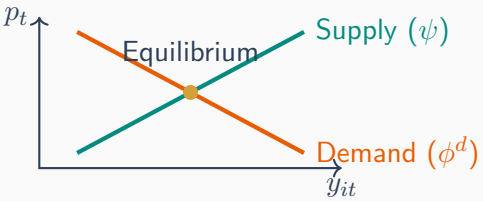


The GIV yields both supply and demand elasticities

Supply Side

$$\psi = \frac{\mathbb{E}[p_t z_t]}{\mathbb{E}[y_{St} z_t]}$$

&

Demand Side

$$\phi^d = \frac{\mathbb{E}[y_{Et} z_t]}{\mathbb{E}[p_t z_t]}$$